

IAN BURKE

Software Engineer · AI Integration & Automation

ianjamesburke@gmail.com ianjamesburke.github.io github.com/ianjamesburke

Software engineer building AI tools for university staff and video production teams.

EXPERIENCE

Davenport University

Current

Automation & AI Integration Specialist

- Built Pounce, a Google Chat agent that helps staff create status reports and donor briefings, submit bugs and ideas, and create Google Docs in shared drives. Built with Next.js, SQLite, Gemini, and locally hosted Gemma and Llama models through LiteLLM.
- Built APIs and MCP servers for university data, including sports search across roughly 282,000 images on campus storage and donor research using Blackbaud NXT. Connected these tools to Pounce, with donor research returned in Google Chat cards and Docs.
- Configured a LiteLLM gateway with separate model keys and spending limits for each app, plus a five-node Gemma pool serving multiple campus agents.
- Built pull-based deployments using systemd and Docker Compose for campus servers that cannot accept inbound connections. Added automated checks to test agent tools after deployment.

Narrative Ads

2022 - Present

Full Stack Developer, NarratorAI

- Built and maintain NarratorAI, an internal platform for AI media production and ads, using TypeScript, Next.js, and Firebase. Brought workflows previously spread across Frame.io, ClickUp, and Google Drive into one application.
- Built image, video, and voiceover generation pipelines using Gemini, fal.ai, ElevenLabs, and OpenRouter.
- Added chat tools that let staff request media generation, review work, and run video-editing jobs.
- Debugged generation jobs, provider failures, and backend capacity issues under load.
- Built tools for querying ad performance data and reconciled monthly costs across GCP, fal.ai, OpenRouter, and Fly.io.

Class Leasing

Spring - Summer 2026

AI Automation Consultant

- Consulted with the admissions and recruiting manager on automating operations for a nationwide team of traveling leasing specialists.
- Built the mockup for a new leasing specialist website, which her team took over and shipped as v1.
- Designed onboarding and offboarding pipelines, plus agent skills and scripts that manage and aggregate flight, rental car, and lodging details into a database agents can query.
- Saved the team about 10 hours a week of manual travel coordination.

SELECTED PROJECTS

Kill Tony Archive – Built a pipeline to transcribe and extract structured data from 1,422 sets across 110 episodes. Used Python, Gemini, SQLite, FastAPI, and SvelteKit.

PLEXI – Built a Rust desktop workspace for coding agents, with terminal panes, command-line agents, and a WASM app runtime. Available as a macOS prerelease.

SKILLS

Python, TypeScript, JavaScript, Rust · React, Next.js, SvelteKit, FastAPI · Firebase, GCP, SQLite, Docker Compose, systemd · LiteLLM, OpenRouter, MCP, agent tool calling, structured extraction, CI/CD